

Hanbo Xie

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<https://xhb120633.github.io/>

Google Scholar: <https://scholar.google.com/citations?user=-ZpfFT8AAAAI&hl=en>

EDUCATION

08/24-Now	Georgia Institute of Technology Atlanta, Georgia, US Ph.D. in Psychology Supervisor: Robert Wilson Minor in Computer Science
03/25-07/25	Princeton University Princeton, New Jersey, US Visiting Student Research Collaborator Supervisor: Tom Griffiths
08/22-05/24	the University of Arizona Tucson, Arizona, US M.A of Psychology Supervisor: Robert Wilson
09/15-06/19	the Southwestern University of Finance and Economics Chengdu, China Bachelor of Management Major: Human Resource Management

EMPLOYMENT

08/24-Now	Georgia Institute of Technology Atlanta, Georgia, US Graduate Assistant Assisting in teaching: <i>AI for Cognitive Science, Social Psychology</i>
08/22-05/24	the University of Arizona Tucson, Arizona, US Graduate Assistant Assisting in teaching: <i>Introduction to Psychology, Personality, Research Methods, Judgment and Decision-Making</i>
07/22-08/22	NeuroMatch Academy Summer School 2022 Global Virtually Project Teaching Assistant Three-week Computational Neuroscience supervision on projects of 8 groups.
07/19-07/22	School of Psychological and Cognitive Sciences, Peking University Beijing, China Research Assistant in Zhou Lab (Supervisor: Xiaolin Zhou) - Assisted a <i>Nature Science Foundation</i> project research (data collection, coding, data analysis, manuscript writing) - Lead an independent research project.

PUBLICATION

* Denotes equal contribution, † Denotes Correspondence, Underscore denotes mentee

Journal Articles:

Qiu, S., Tang, Y., Yu, H., **Xie, H.**, Dreher, J. C., Hu, Y., & Zhou, X. (2025). Toward a computational understanding of bribe - taking behavior. *Annals of the New York Academy of Sciences*.

Fang, Z., Zhao, M., Xu, T., Li, Y., **Xie, H.**, Quan, P., ... & Zhang, R. Y. (2024). Individuals with anxiety and depression use atypical decision strategies in an uncertain world. *eLife*, 13.

Xie, H. (2023). The promising future of cognitive science and artificial intelligence. *Nat Rev Psychology*.

Conference:

Zhu, J.-Q.*, **Xie, H***, Arumugam, D., Wilson, R. C., & Griffiths, T. L. (2025). *Using reinforcement learning to train large*

language models to explain human decisions. arXiv preprint arXiv: arXiv:2505.11614. ICLR 2026.

Pan, L.*, Xie, H*†., & Wilson, R. C. (2025). Large Language Models Think Too Fast To Explore Effectively. arXiv preprint arXiv:2501.18009. NeurIPS 2025 Poster.

Xie, H*†., Zhu, J. Q., Xiong, H. D., Wilson, R., & Griffiths, T. (2025). Reasoning Across Minds and Machines. In *Proceedings of the Annual Meeting of the Cognitive Science Society* (Vol. 47).

Zhang, Z.*, Xie, H*, Baker, T., Peters, M., & Wilson, R. C. (2025). Linking strategies to think aloud in a stochastic learning task. In *Proceedings of the Annual Meeting of the Cognitive Science Society*.

Xie, H., Xiong, H., & Wilson, R. C. (2024) From Strategic Narratives to Code-Like Cognitive Models: An LLM-Based Approach in A Sorting Task. *First Conference on Language Modeling (COLM)*.

Xie, H., Xiong, H., & Wilson, R. C. (2024) Evaluating Predictive Performance and Learning Efficiency of Large Language Models with Think Aloud in Risky Decision Making. *Computational Cognitive Neuroscience (CCN), MIT*.

Xie, H., Xiong, H., & Wilson, R. C. (2023). Text2Decision: Decoding Latent Variables in Risky Decision Making from Think Aloud Text. *NeurIPS 2023 AI for Science Workshop*.

Xie, H., Xiong, H., & Wilson, R. C. (2023). Computational introspection: Can large language models reveal cognitive algorithms from human language? Poster session presented at the *5th Chinese Computational and Cognitive Neuroscience Conference*, Beijing, China.

Guo, Y., Song, S., Xie, H., Gao, X., & Zhang, J. (2022, February). ARIMA and RNN for Selection Sequences Prediction in Iowa Gambling Task. In *2022 2nd International Conference on Artificial Intelligence and Signal Processing (AISP)* (pp. 1-6). IEEE.

Song, S*, Xie, H*, Speekenbrink, M., Zhang, J., Gao, X., & Zhou, X. (2020, October). The computational basis of individuals' learning under uncertainty in groups with collective goals. Oral presentation at the Society for Neuroeconomics, Vancouver, Canada.

Preprints and Submitted Works:

Xie, H*, & Zhu, J*. (2025, July 12). Centaur May Have Learned a Shortcut that Explains Away Psychological Tasks. https://doi.org/10.31234/osf.io/u7z4t_v1 (submitted).

Xie, H*†., Xiong, H. D., & Wilson, R. C. (2025) Rethinking Think-Aloud in the Age of Language Models. PsyArXiv. https://osf.io/preprints/psyarxiv/6ta3z_v1 (submitted)

Xie, H*†., Jagadish, A. K., Pan, L., & Wilson, R. C. (2026). Think-aloud reshapes automated cognitive model discovery beyond behavior. arXiv. <https://doi.org/10.48550/arXiv.2605.05091> (submitted)

FUNDING & GRANTS

- OpenAI Researcher Access Program (2024-2025) "What does think aloud reveal on human cognitive process?" (\$5,000)

CONTESTS & AWARDS

- The 1st Chinese Computational Psychiatry Hack : **The Champion Team**
- Better Together Psychology Conference: **Best Flash Talk Award**

LECTURES & INVITED TALKS

- **Understanding Human Thoughts from Think Aloud: An LLM Approach (Invited Talks)**
Invited by Fudan Institute of Science and Technology for Brain-Inspired Intelligence (Lab PI: Dr. Tianye Jia), School of Psychological and Cognitive Sciences, East Normal University (Lab PI: Dr. Xiaolin Zhou), Institute of Neuroscience, CAS (Lab PI: Dr. Tianming Yang), School of Psychological and Cognitive Sciences, Peking University (Lab PI: Dr. Hang Zhang). (May - June, 2024).
- **From Behavior to Minds: An Overview of Computational Modeling in Psychology (Invited talk)**
Invited by Research Methods Community (Jan, 2024).

- **Forging the Future: Uniting AI and Cognitive Science through Large Language Models (Invited talk)**
Invited by School of Psychological and Cognitive Sciences, East Normal University (May, 2023).
- **Computational Modeling Basics and Implementations in Stan. (Invited talk)**
Invited by Institute of Applied Psychology in Tianjin University (Sep, 20, 2021, codes and slides available on GitHub)
- **Reinforcement learning in computational cognitive sciences. (Lecture)**
Video available at: [Reinforcement Learning Modeling Club 1 bilibili](#) (in Chinese, codes and slides available on GitHub)

COMMUNITY

- 08/2025: Leading co-organizer of workshop at CogSci 2025: Reasoning Across Minds and Machines (https://xhb120633.github.io/reasoning_workshop/)
- 07/2023: Co-Founder of MindRL Hub (<https://rldmjc.github.io/>) (A global community for Reinforcement Learning scholars from Psychology, Neuroscience and AI)
- 2022 Fall: Mentor in ASFP (for graduate applications mentoring)
- 2022 Fall: Mentor in PRP (for mentoring undergraduate and master students to do research)

SKILLS

- Programming Skills: Matlab (Psychtoolbox, MLE modeling, VBA toolbox), R(Visualization, RStan), Python (NumPy, psychoPy, PyTorch), JavaScript (JSPsych), CSS, Markdown, LaTeX
- Analytical Skills: Statistical analysis, Computational Modeling(non-Bayesian and Bayesian Modeling, machine learning tools, Neural Networks, fMRI data analysis, Large Language Models(distributed training/inference, mechanism analysis).

AD-HOC REBIEWS

- Journals: *AMPPS*, *Computational Psychiatry*, *Scientific Reports*
- Conferences: *AAAI*, *Annual Meetings of Cognitive Sciences (CogSci)*, *Computational Cognitive Neuroscience (CCN)*, *COLM*, *ICML*, *ICLR*, *NeurIPS*